

Development of RFID Enabled Aircraft Maintenance System

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Abstract—Recently, US Federal Aviation Administration (FAA) have formally approved the use of passive radio frequency identification tags on individual airline parts. Air bus and Boeing proposed amending the ATA's Specification 2000 to include the use of ISO 18000-3 (13.56MHz) and 18000-6 (UHF). They also announced plans to develop a single RFID specification to be used by both companies to use RFID tags to identify commercial aircraft parts. This decision will drive a wide adoption of RFID technology in aviation industry. In this paper, we present a RFID enabled aircraft maintenance system which could be integrated with inventory control system and aircraft scheduling system to ensure on time performance and safety of passenger and cargo. We believe our study would give a good guideline for further researches on RFID application in the aviation industry and aircraft part manufacturing industry.

Index Terms—Aviation, ATA, RFID, Maintenance System.

I. INTRODUCTION

In 2006, IDTechEx expect 1.3 billion tags to be sold, with 500million RFID smart labels for pallet and case level tagging but the majority into a range of diverse markets from baggage and passports to contactless payment cards and drugs (<http://www.idtechex.com>). They also see that the value of the total market including systems and services will jump from \$2.71Bn in 2006 to \$26.23Bn in 2016. As in the

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many reports, there is no doubt about the growth of RFID's future market.

In aviation industry, major airports have been looking for the opportunities in baggage handling area since 1999. Many pilot tests have been done at numerous U.S. and European airports including Gimpo, Las Vegas, Jacksonville, Seattle, Los Angeles, San Francisco, Heathrow, Boston, New York and Rome. In the U.S. tests, RFID tags were far more accurate than bar codes when applied to baggage handling operation. According to ICAO report (<http://www.icao.int>), nearly every case (and certainly with all of the most mature, more recently tested RFID baggage tag systems) RFID performance was measured with read accuracies well above 98 per cent, whereas similar barcode baggage tag read accuracies generally fell below 85 per cent

Boeing has recently announced that suppliers of many parts used in its 787 Dreamliner will be required to place RFID tags on the parts before shipping them to Boeing [1]. It is expected that hundreds of Boeing suppliers and sub-suppliers will need to begin tagging parts with passive UHF tags. Together with Boeing, Airbus is also promoting the adoption of industry solutions for RFID on commercial aircraft parts. They believe that RFID could provide major benefits for the entire industry. They will get more accurate information about their demand for parts and will be able to reduce their parts inventory and cut the time it takes to repair planes. Part suppliers will also be able to reduce inventory, improve the efficiency of their manufacturing operations and reducing the amount of unapproved parts that enter the supply chain [2].

In this paper, we will show a RFID enabled aircraft maintenance system, and also show RFID enabled part management system. We believe our study would give a good guideline for further researches on RFID application in the aviation industry and aircraft part manufacturing industry

II. CONVENTIONAL AICRAFT MAINTENANCE

Maintenance management maybe defined as all activities required to keep equipment to a specified operating condition. The activities involved in the maintenance should be conducted with a cost efficient way and with safe

environment. Since 1980, many companies have introduced maintenance management information systems for their maintenance and the role of IT on maintenance considered to be expanded as the service requirements become more specific and complex [3]. Also due to the fast growth of IT market and technology itself, the evolution from tele-maintenance (or tele-service) to e-maintenance solutions (or wireless solutions) are rapidly emerged [4, 5]. In aviation industry, aircraft maintenance system can enhance safety of passenger/goods, ensure on time performance and increase asset availability.

In the paper, we surveyed maintenance systems in several airlines in Asia. We found that there were difference among companies, but conventionally maintenance system consists of maintenance head-quarter (MH), part inventory area (PI), maintenance line (ML), flight schedule department (FS), part repair shop (PR) and part manufacturer/supplier (PM). Fig. 1 and 2 show a conventional aircraft maintenance process flow among different areas and an activity diagram, respectively.

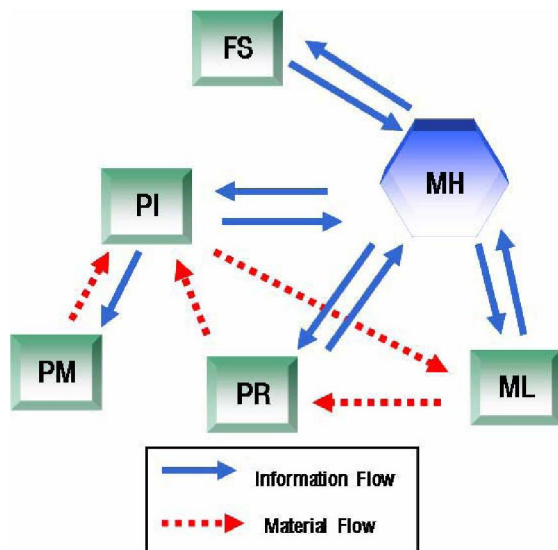


Fig. 1. Aircraft maintenance process flow

The role of each area is as follows:

- MH is managing the aircraft maintenance and builds a maintenance schedule based on the regulation (e.g. FAA regulation and airline regulation) on the part. It also send the schedule to ML (and also to PI) to discuss manpower and part requirement; build schedules to send parts to PR in case local maintenance is impossible; send maintenance schedule to FS in order to help them to plan flight schedules.
- ML builds a detail schedule based on the maintenance schedule build by MH. When some part maintenance is impossible in local maintenance shop, it send the order to PR which is in the other airport (or sometimes abroad).
- PI is a part warehouse to keep inventory.

- FS is scheduling department for flight operation. Flight schedule is based on the maintenance schedule (usually scheduled maintenance).
- PR is repairing parts requested by MH. Once part is repaired it sends the part to PI.
- PM supports parts to PI requested by PI.

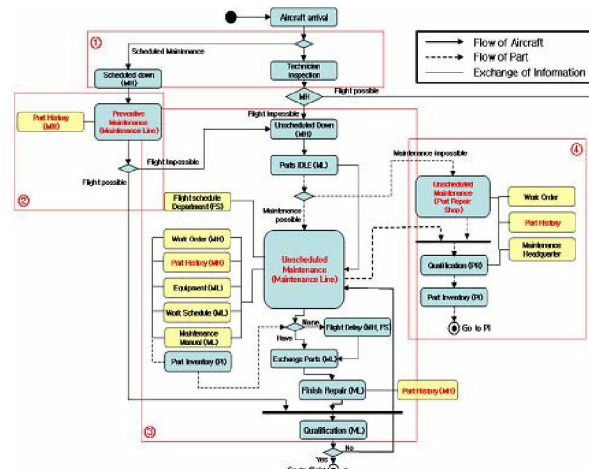


Fig. 2. Aircraft maintenance activity diagram

In the Fig. 2, description of each block is as follows:

- ① Once aircraft is arrived, maintenance schedule is reviewed. If the aircraft has maintenance schedule, the information is sent to MH to prepare maintenance. If the state of the aircraft is not normal, the aircraft status is changed to unscheduled maintenance and the information is sent to MH for further action even if there is no schedule for maintenance.
- ② is preventive maintenance area. After preventive maintenance, if the aircraft is operational, the aircraft is ready for flight. If the status of aircraft is not operational, the aircraft should go to unscheduled maintenance process.
- ③ is ML area (unscheduled maintenance area).
- ④ is a process for some parts which are sent to the other airport (or abroad). PR is receiving parts for repair from ML, and sends those parts to PI after repair.

As in the Fig. 1, and Fig. 2, conventional system is heavily dependent on the MH. The system has also following issues:

- It has difficulty in tracking parts (no visibility).
- Manual input of data
- Manual inventory control
- It is hard to trace maintenance history of part.
- No integration with ERP, flight operation schedule

III. RFID ENABLED MAINTENANCE

Survey on maintenance and inventory procedures in the airline and maintenance organization indicated that 87% of respondents were using Reorder Point (ROP) system and 13% were using Material Requirement Planning (MRP) system [6]. However, it was seen that about half of the

companies who were using the ROP were dissatisfied and considering implementing the MRP system.

A new approach on the application of unitary software structure composed of the critical path method (CPM), material requirement planning (MRP), and production activity control (PAC) techniques was tried by Australian airline [7]. The reason behind this approach was that there were many companies which were equipped with CPM, MRP and PAC, but these systems were somehow isolated. Even unitary structure, encompassing BOM, CPM and operation routing, still required further enhancement to accommodate this situation in maintenance scheduling. In order to address the inefficiency in data handling, we designed aircraft maintenance data warehouse as in Fig. 3, changed the process.

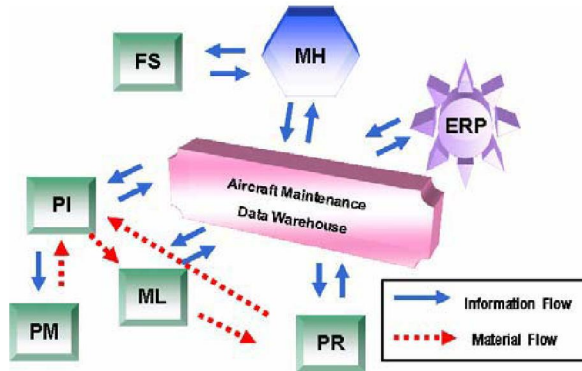


Fig. 3. Aircraft maintenance data warehouse and new maintenance process.

To address visibility issue and data input issue, we reengineered the existing maintenance process and developed prototype RFID based system (Fig. 4).

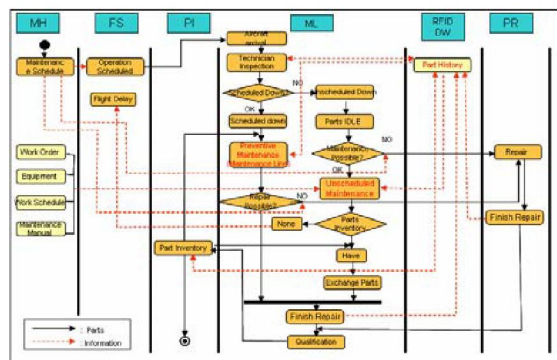


Fig. 4. Activity diagram of RFID based maintenance

As in the figure, RFID data warehouse is managing data for maintenance schedule. It collects part information (status, history, etc) and provides service for required part for maintenance.

Fig. 5 shows master record data in the prototype system.

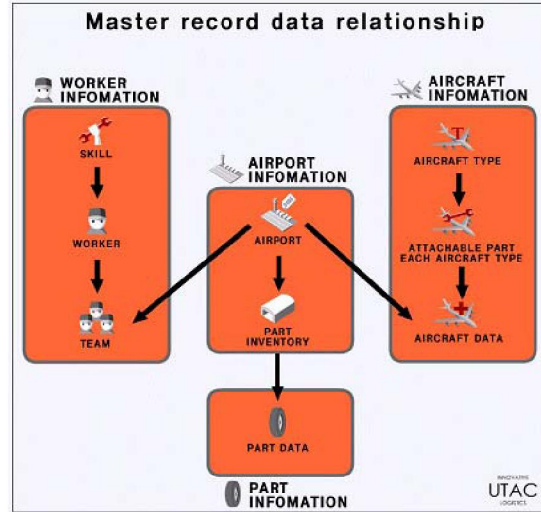


Fig.5 . Data architecture for prototype system

An interesting concept on the maintenance, repair and overhaul (MRO) of aircraft, was introduced in order to make MRO more efficient [8]. However, the proposed system only shows the potentials of ubiquitous environment on the maintenance, repair and overhaul operation (MRO) but did not shows the picture of truly integrated environment among aircraft operation system, maintenance system and inventory management system which work tightly together.

IV. SERVICE SCENARIOS

A. Service procedure

Fig. 6 shows a brief data base structure for testing our prototype system. In this scenario, we focused on the tire maintenance case.

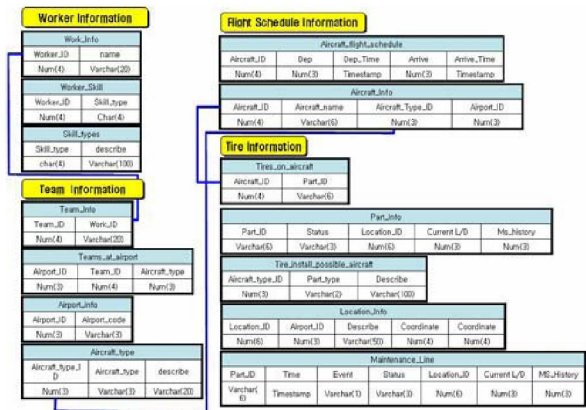


Fig. 6. Data base structure example

In the database, ML consists of team related tables and worker related tables at each airport. In the team related tables, team structure at each airport and aircraft type (i.e. available aircraft type for maintenance) at each airport were shown. Worker table describes the skills of each worker and relevant descriptions. Since flight schedule is dependent on the aircraft maintenance schedule, aircraft should terminate its operation and be in the maintenance task at the pre-planned airport.

The following scenario describes aircraft maintenance processes using RFID technologies. In this scenario each technician (i.e. worker) has a personal digital assistant (PDA) or mobile phone as a means of communication. Fig 8 shows the procedure to generate a maintenance order. The following sequence explains a service scenario for aircraft maintenance.

- Maintenance headquarter decides the maintenance of a specific aircraft using information in data warehouse (Based on this plan, ML and PI start their tasks).
- After comparison with BOM file with aircraft type, a specific aircraft BOM file is chosen for explosion (in the figure “X” means not matched or not chosen).
- An airport for maintenance is decided based on the status of part assignment as well as worker assignment.
- After checking part inventory and available skilled workers, the maintenance head quarter decides specific airport for maintenance and generates a maintenance order (here, part inventory status is checked with the aid of RFID based inventory control system).
- Workers receive aircraft information from data warehouse that is going to enter the maintenance shop and also receive work order list (e.g. in the work order list, there can be data such as aircraft number, action code, action time, parts ID).
- If there are parts need be replaced, those parts should be prepared (or requested) timely for maintenance.
- An aircraft is entering to the maintenance line according to the maintenance schedule.
- The assigned workers are doing maintenance. Workers can use PDA to see maintenance manual.
- One can check maintenance history of part from maintenance data warehouse.
- During the maintenance tasks, maintenance activities are update from maintenance line.
- Once maintenance is finished, workers send maintenance report to the maintenance data warehouse.

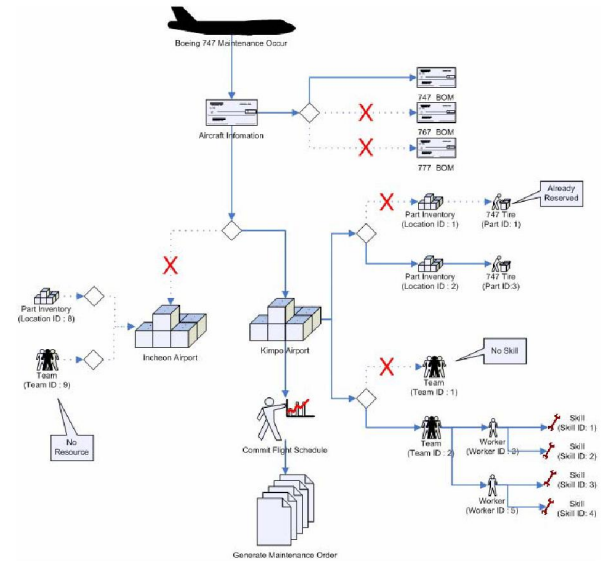


Fig 8. Generation of maintenance order

Fig 9 shows an example of data manager in the data warehouse (setting up a maintenance team). As in the figure, each worker can have more than 1 skill and a team consists of skilled worker for specific flight type. Each airport consists of team whose skill set could be different from the other airport (i.e. the maintenance of an aircraft may be only possible at specific airport due to constraint of resource availability and part availability. Flight operation schedule should consider the maintenance team located at each airport).

Worker ID	Name	Skill
41	Oh chang	MT01, MT03
42	Yoon sun	MT02
43	Sung Bae	MT03
44	Sung min	MT01, MT02
45	Vijai	MT02, MT03
46	Jang Ahn	MT01, MT02, MT03
47	Vijendra	MT02

Aircraft Type	Description
747	Boeing 747
757	Boeing 757
767	Boeing 767
777	Boeing 777

Location	Aircraft	Team ID	Workers(ID-Name)	Command
GMP	747	1	41(Oh chang), 42(Yoon sun)	Modify Delete
GMP	777	7	43(Sung Bae), 46(Jang Ahn)	Modify Delete

Fig 9. Maintenance team set up

Fig 10 is a snapshot of maintenance head quarter which shows the status of some part (here, tire) based on the usage.

“Red” means a status of part, which requires “urgent maintenance”, “yellow” means a status of part, which needs “attention” and “green” means a status of part, which is “in good states”.

Status	Current Landing	Aircraft	Part ID	Part Type	Current Location	Command
Red	368	HL0002 (757)	21	T1	GMP	MS Add
Red	368	HL0002 (757)	23	T1	GMP	MS Add
Red	366	HL0001 (747)	26	T5	GMP	MS Add
Yellow	365	HL0001 (747)	7	T6	GMP	MS Add
Yellow	357	HL0001 (747)	25	T4	GMP	MS Add
Yellow	355	HL0001 (747)	13	T4	GMP	MS Add
Green	4	HL0002 (757)	14	T1	GMP	MS Add
Green	3	HL0001 (747)	5	T2	GMP	MS Add
Green	2	HL0001 (747)	9	T6	GMP	MS Add
Green	1	HL0005 (747)	6	T4	GMP	MS Add
Green	1	HL0004 (777)	8	T1	GMP	MS Add

Fig 10. Maintenance head quarter

B. Scenario for inventory control process with intelligent shelf system (Fig 11)

- Ordered parts are arrived at the warehouse or stock location (Using gate RFID reader, part information is entered).
- Store part in the designate shelf (In here, we used shelf antennas).
- Smart shelf system will help to improve putaway process by considering stock location.
- Once stored, intelligent shelf system checks part information in real time.
- System sends tag-scan information to the system when there is change (Analyze part inventory status). Also it can send SMS message to the operator or inventory controller when necessary.
- When necessary system generates procurement suggestion based on the EOQ and ROP (but purchase order is generated by ERP and system only generates procurement suggestion).
- ERP will generate purchase order suggestion.

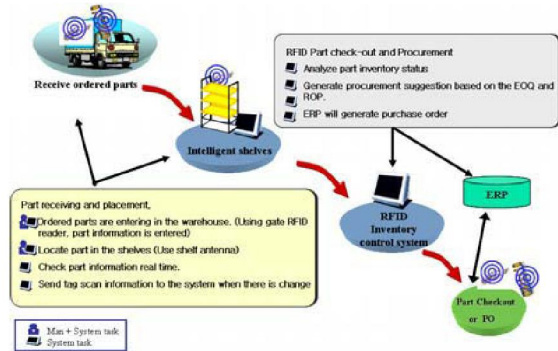


Fig 11. Inventory management

Fig 12 shows an event handling example when parts are misplaced in the shelf. Based on the store location information and part data information, the system checks and compare the location of parts. In aviation industry, it is well known that technicians generally spend over 20% of their time to search for parts/tools and the proposed capability can drastically reduce such loss. The picking process is heavily dependent on the putaway process. In order to reduce the picking process time and improve putaway accuracy, this kind of event handling process was designed and suggested.

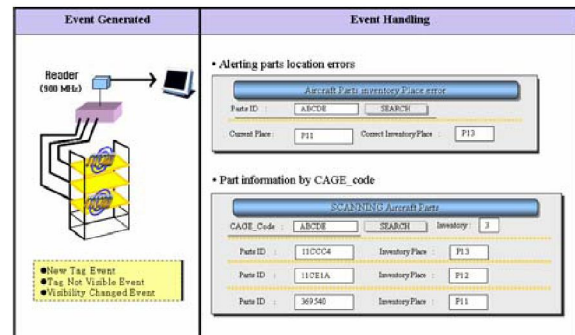


Fig 12. Intelligent shelf and part management system

C. Technical Requirement for RFID application

For successful adoption of the concept developed, the following requirement should be satisfied:

- Read reliability and accuracy considering distances
- Reading reliability and accuracy considering antenna type (e.g. circular, rectangular) and antenna location.
- Reading reliability and accuracy considering tag orientation and angle.
- Writing distance
- Impact of multi-readers environment
- Impact of material on reading capability (e.g. distance)
- Network requirements

on RFID application in the aviation industry and aircraft part manufacturing industry.

D. Environmental Requirement for RFID application

Environmental effects to collect data (e.g. tag identification) are as follows:

- Effect from aviation equipment or guidance system or ETV (Elevating Transfer Vehicle) etc.
- Effect from other wireless device in the same area
- Effect of environment (temperature, humidity, shock etc)
- Effect of power breakdown, fire, etc

E. Potential benefits of prototype system

Potential benefits of our prototype system are summarized in the Table 1. As in the table, it is considered that the proposed system will help to achieve more efficient aircraft maintenance process by providing more reliable information on the maintenance floor and storage information in the warehouse.

Table 1. Potential benefit of the proposed system

Existing System	Proposed RFID Based System
Incorrect part data and delayed data update due to non-real time based data collection	Real time update of part data
Human error when entering part information	Removal of Human error
No integration among different module (e.g. maintenance headquarter, maintenance line, part inventory, etc)	Integration throughout data warehouse among different modules
Unable to manage product lifecycle and hard to trace part history.	Able to manage product lifecycle throughout its lifecycle
Hard to support technician remotely.	Able to support, technician remotely

V. CONCLUSIONS

In this paper, we showed prototype RFID enabled aircraft maintenance system which was integrated with RFID enabled part management system. The proposed system consists of prototype systems developed in the laboratory environment. From our study, we found the following issues: Current aircraft component (i.e. part) are not designed for RFID tagging; RF tag packaging should be designed for aircraft component; Part numbering scheme should be developed in order to adopt EPC numbering scheme or other standard numbering scheme; In order to meet required performance level, UHF tags should not be attached directly to the metal component, or metal tags should be used.

Even though, the systems were developed and implemented under laboratory environment, we believe that our study would give a good guideline for further researches

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